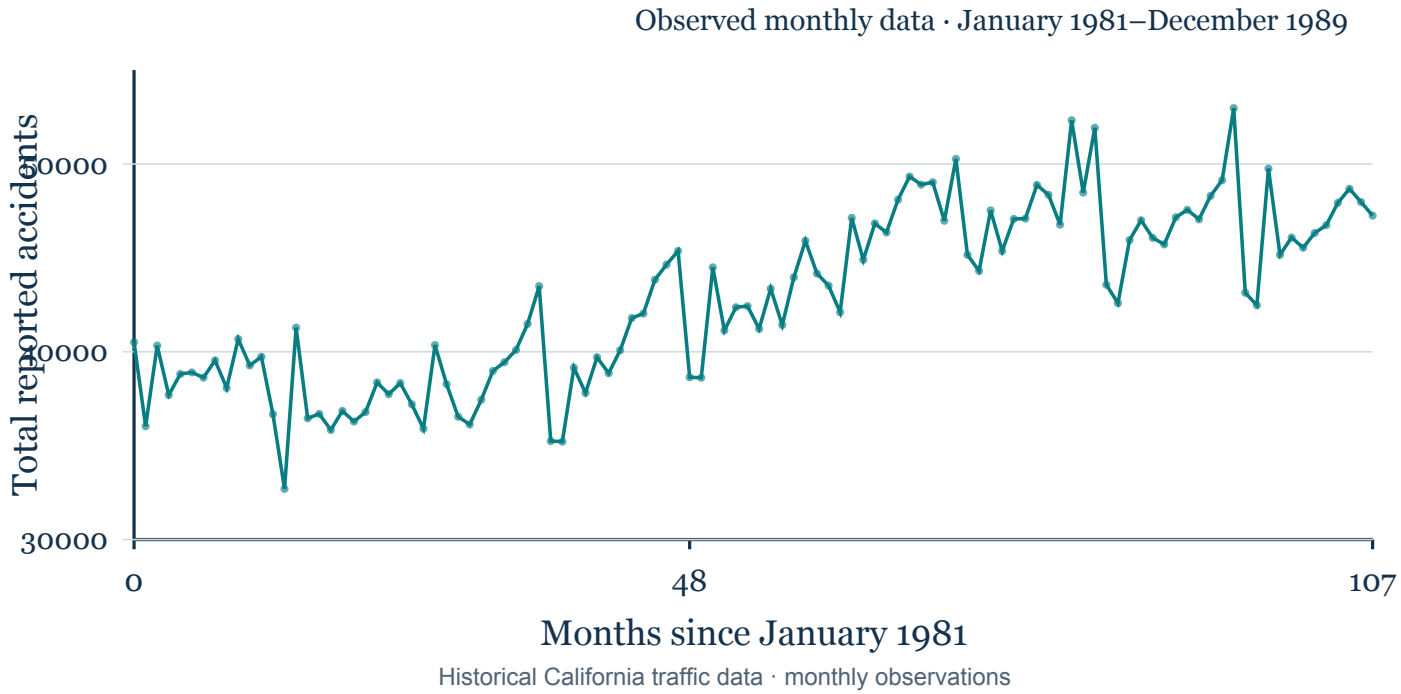
Did a traffic law change accidents, or did the calendar change?

Begin with a dated outcome and a plausible alternative explanation.



Recall the problem of a time-only policy variable

Time controls can remove useful variation as well as confounding.

> The previous block showed that unrestricted year indicators absorb any variable with one common value in each year.

> We therefore need to distinguish seasonal indicators, which repeat by calendar month, from an indicator for every unique month-year.

Define the monthly unit and time index

A lag requires a correct ordering, not just a row number.

> $t = 1, \ldots, 108$: monthly observations

> January 1981 through December 1989

A trend is a gradual time pattern

In a log outcome, a linear trend describes proportional growth.

> $\log A_t = a + gt + u_t$

> Small-change interpretation: about $100g\%$ per month

Seasonality repeats at the same part of the year

Use a reference month to avoid redundant indicators.

> January is the reference category

> Eleven recurring month indicators, not 108 unique date indicators

Frequency and season

Apply the idea before reading the solution.

Practice

An accident dataset runs monthly from January 1981 through December 1989. How many observations are there, and why use eleven month indicators with an intercept?

Frequency and season: explanation

Connect the calculation to the empirical claim.

> There are $9 \times 12 = 108$ observations. Eleven indicators compare months with an omitted reference month; including all twelve plus an intercept creates exact collinearity. Their purpose is seasonal adjustment, not causal identification by itself.

Write the policy model with its calendar controls

Make the intervention and units explicit.

> $\log A_t = a + gt + \sum_{m=2}^{12} \gamma_m M_{mt} + bB_t + cS_t + dU_t + fW_t + u_t$

> M_{mt}: month indicator; γ_m: shift relative to January

Inspect the calendar-only regression

Understand the baseline before adding policy variables.

```
. regress logacc trend i.month

------------------------------------------------------------------------------
      logacc | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       trend |   .0027471   .0001611    17.06   0.000     .0024274    .0030669
             |
       _cons |   10.46857   .0190028   550.90   0.000     10.43084    10.50629
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

A seasonal joint test asks about the whole pattern

One insignificant month does not settle the annual question.

```
. testparm i.month

 ( 1)  2.month = 0
 ( 2)  3.month = 0
 ( 3)  4.month = 0
 ( 4)  5.month = 0
 ( 5)  6.month = 0
 ( 6)  7.month = 0
 ( 7)  8.month = 0
 ( 8)  9.month = 0
 ( 9)  10.month = 0
 (10)  11.month = 0
 (11)  12.month = 0

       F( 11,    91) =   10.29
            Prob > F =    0.0000
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

Contemporaneous exogeneity concerns the current regressor vector

Current means the variables included for the same date.

> $E(u_t \mid x_t) = 0$

Strict and sequential exogeneity condition on different information

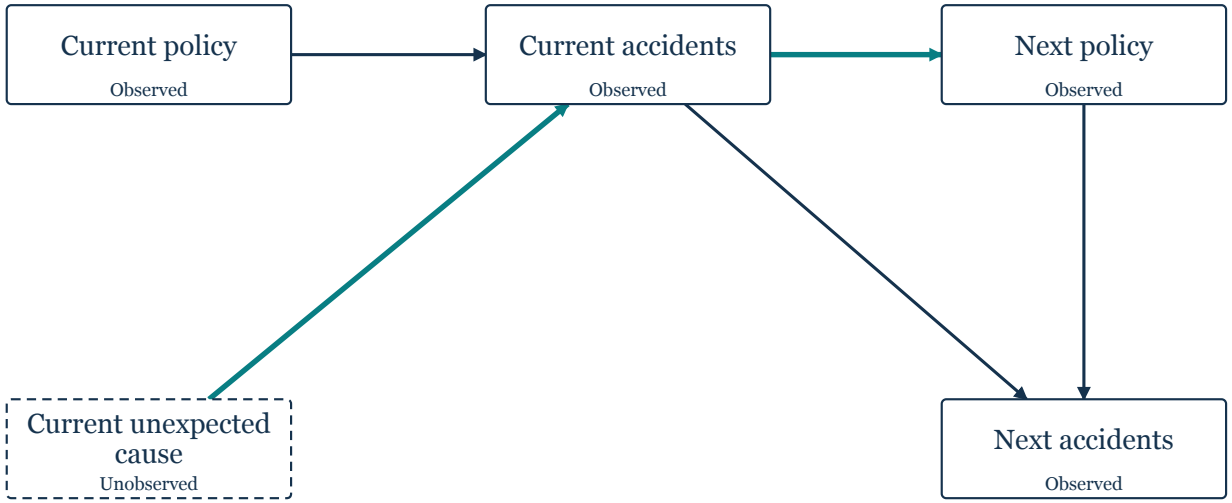
Future regressors make strict exogeneity demanding.

> Strict: $E(u_t \mid x_1, \ldots, x_T) = 0$

> Sequential: $E(u_t \mid x_1, \ldots, x_t) = 0$

Feedback across dates does not require a causal cycle

Current accidents can affect the policy chosen next month.



1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Possible route: $u_t \to Y_t \to X_{t+1}$
- Future policy can respond to current unexpected outcomes.

Past outcomes can contain past errors

Adding dynamics changes the exogeneity question.

> A lagged outcome is the outcome from an earlier date.

> This is a reason to model timing explicitly rather than assume lagging a variable always makes it safe.

Strict exogeneity

Apply the idea before reading the solution.

Practice

Define strict and contemporaneous exogeneity for a time-series regression using x_t and error u_t. Which is stronger?

Strict exogeneity: explanation

Connect the calculation to the empirical claim.

> Contemporaneous exogeneity requires $E(u_t \mid x_t) = 0$. Strict exogeneity requires $E(u_t \mid x_1, \ldots, x_T) = 0$ for every t, including future regressors. Strict implies contemporaneous, but the reverse need not hold. These statements use x_t as the entire regressor vector.

Read the adjusted accident regression

Connect the coefficient with the actual outcome and controls.

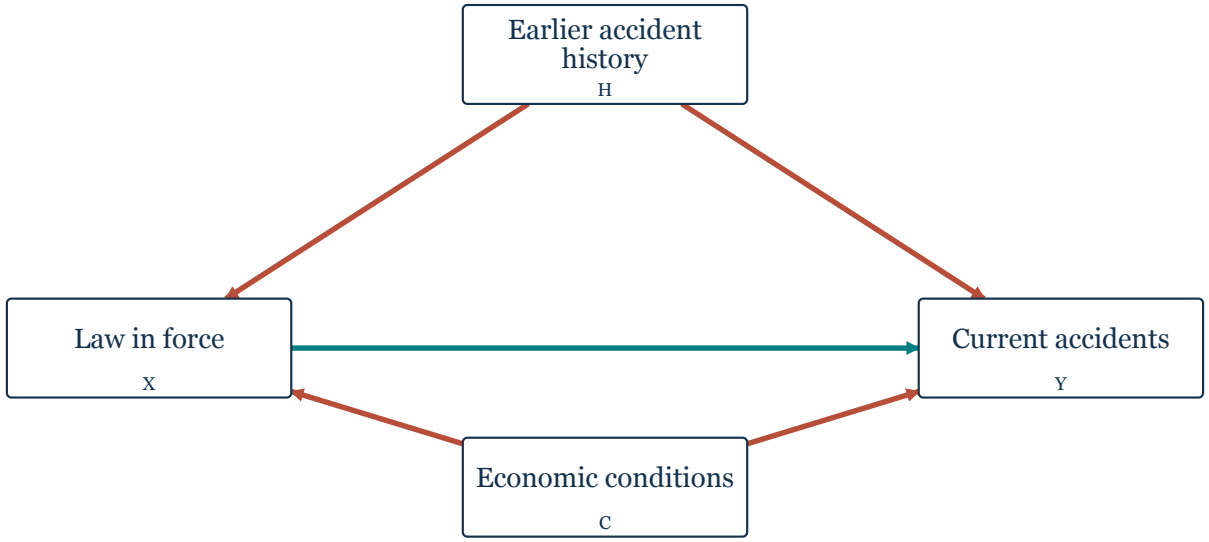
```
. regress logacc trend i.month unem spdlaw beltlaw wkends, vce(robust)

------------------------------------------------------------------------------
             |               Robust
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       trend |   .0011011   .0002676     4.12   0.000     .0005696    .0016326
             |
        unem |  -.0212173   .0038238    -5.55   0.000    -.0288128   -.0136218
      spdlaw |  -.0537593   .0120059    -4.48   0.000    -.0776076    -.029911
     beltlaw |   .0954526   .0123682     7.72   0.000     .0708847    .1200206
      wkends |   .0033333   .0039443     0.85   0.400    -.0045016    .0111682
       _cons |   10.63986   .0681238   156.18   0.000     10.50454    10.77518
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

A date indicator can mix several simultaneous changes

A coefficient needs a counterfactual interpretation.



1 open directed route · 2 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

A count and a rate answer different questions

Exposure can change while individual risk stays fixed.

> Total accidents can rise because there is more driving, even if risk per trip falls.

> Our current outcome is the recorded statewide count, logged for the fitted comparison.

Time ordering changes the meaning of a sample

One realised history can contain strongly related observations.

> A cross-sectional sampling argument often imagines independent draws of units.

> The law of large numbers and central limit reasoning used for independent observations require suitable extensions for dependent histories.

A linear trend does not remove every nonstationary process

Distinguish a chosen path from cumulative random shocks.

> A deterministic trend is a specified function of time.

> Before interpreting relationships among trending variables, inspect their time plots and consider a model for their persistence.

Record the time-series specification in words

Dates and frequency are part of reproducibility.

> A useful description gives the date range, monthly frequency, outcome scale, policy coding, trend and seasonal controls.

> Then explain the proposed economic comparison and its main uncertainty and identification conditions.

Before–after is not enough

Apply the idea before reading the solution.

Practice

Accidents fall after a road-safety law. Why is this not yet evidence of its causal effect?

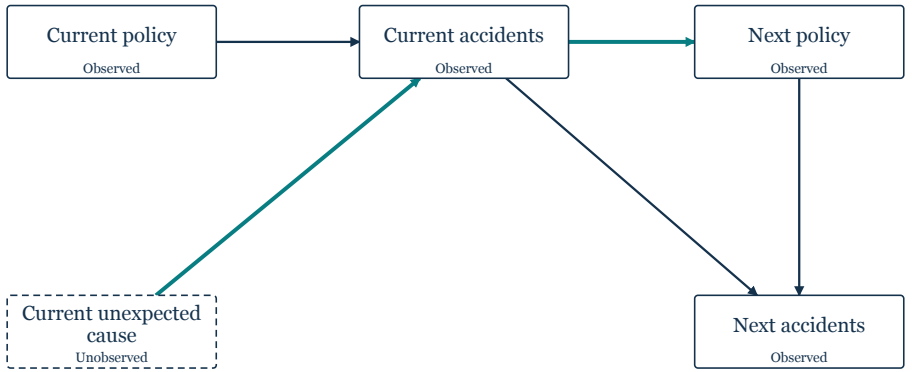
Before–after is not enough: explanation

Connect the calculation to the empirical claim.

> Traffic volume, seasonality, fuel prices, enforcement, reporting or other policies may also change. A defensible causal comparison must rule out relevant counterfactual changes; adding a trend is useful only insofar as its assumptions are credible.

DAG practice: dated policy feedback

Trace the routes before choosing a control.



1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Current unexpected accident causes can affect current accidents and hence next month's policy. Trace that dated route. Why can it challenge strict exogeneity without requiring a future policy to cause an earlier accident?

DAG practice: dated policy feedback: explanation

Explain what conditioning changes in this stated graph.

> The route is $u_t \to Y_t \to X_{t+1}$. It provides a reason that the current disturbance can be related to future regressors, which strict exogeneity conditions on. Every causal arrow still moves forward through the policy process. A sequential mean assumption uses a different information set and can be considered when current policy is chosen before the current unexpected cause, subject to the full model and timing assumptions. The graph does not prove that assumption. It makes clear why the strict version should not be imposed merely because the observations have dates.

Interpret the traffic-law coefficient with its timing

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_09.do using traffic.csv. Verify the monthly date range and sample size. Translate the seat-belt-law coefficient into an exact percentage contrast in exponentiated fitted log accidents. Explain why the trend and month controls do not establish that the law caused this contrast.

Interpret the traffic-law coefficient with its timing: explanation

A reproducible result needs both a calculation and an interpretation.

> The series has 108 months, January 1981 through December 1989. Its seat-belt-law coefficient is 0.095453. The exact fitted multiplicative contrast is 10.0157 percent. It concerns total accidents, rather than deaths or accidents per distance travelled. The result compares law states after accounting for the specified trend, seasonal pattern and other included variables. Adoption could coincide with unmeasured enforcement, reporting, exposure or other policy changes. Controlling for the calendar addresses the pattern represented by those controls. A causal statement needs a credible argument about the remaining counterfactual accident path, rather than relying on the law indicator's p-value.

Next, ask how long a shock lasts

Persistence determines useful dynamics and long-run comparisons.

> We have separated trend, recurring seasonality and remaining errors, and we have defined exogeneity with its time information.

> We will use autoregressive and distributed-lag models to make those possibilities explicit.